

POWER PLANT ECONOMICS

Load Curves

- The curve showing the variation of load on the power station with respect to (w.r.t) time is known as a load curve.
- These load variations during the whole day (i.e., 24 hours) are recorded half-hourly or hourly and are plotted against time on the graph. The curve thus obtained is known as daily load curve as it shows the variations of load w.r.t. time during the day

Daily Load Curve-1



Daily Load Curve-2

- (i) The daily load curve shows the variations of load on the power station during different hours of the day.
- (ii) The area under the daily load curve gives the number of units generated in the day.
Units generated/day = Area (in kWh) under daily load curve.
- (iii) The highest point on the daily load curve represents the maximum demand on the station on that day.
- (iv) The area under the daily load curve divided by the total number of hours gives the average load on the station in the day.

$$\text{Average load} = \frac{\text{Area (in kWh) under daily load curve}}{24 \text{ hours}}$$

- (v) The ratio of the area under the load curve to the total area of rectangle in which it is contained gives the load factor.

$$\begin{aligned} \text{Load factor} &= \frac{\text{Average load}}{\text{Max. demand}} = \frac{\text{Average load} \times 24}{\text{Max. demand} \times 24} \\ &= \frac{\text{Area (in kWh) under daily load curve}}{\text{Total area of rectangle in which the load curve is contained}} \end{aligned}$$

- (vi) The load curve helps in selecting* the size and number of generating units.
- (vii) The load curve helps in preparing the operation schedule** of the station.

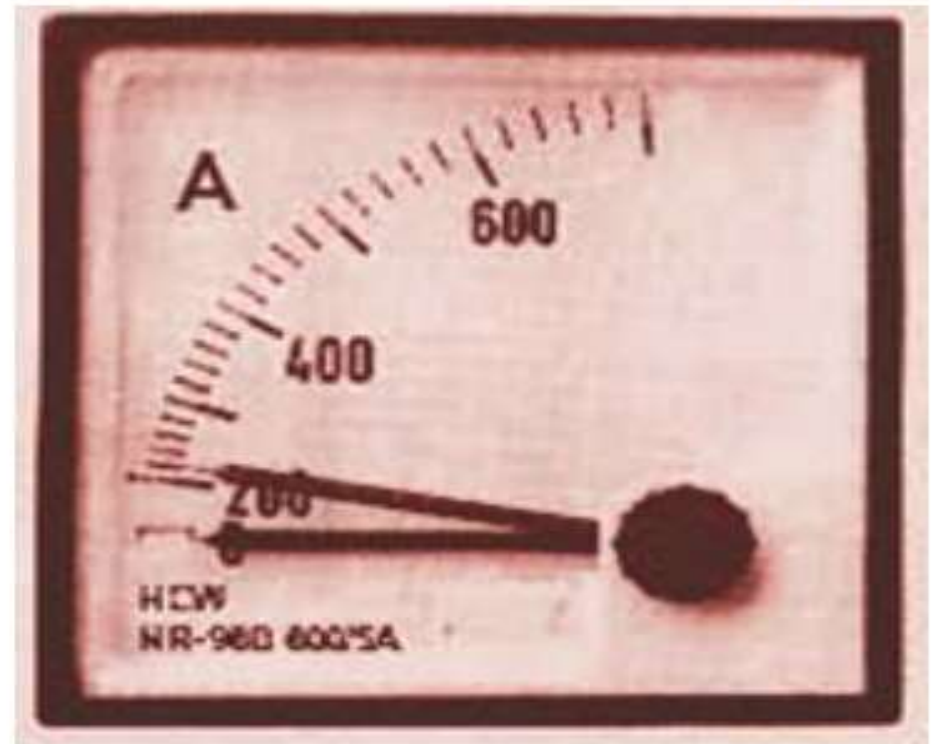
Important Terms

(i) Connected load. *It is the sum of continuous ratings of all the equipments connected to supply system.*

A power station supplies load to thousands of consumers. Each consumer has certain equipment installed in his premises. The sum of the continuous ratings of all the equipments in the consumer's premises is the "connected load" of the consumer. For instance, if a consumer has connections of five 100-watt lamps and a power point of 500 watts, then connected load of the consumer is $5 \times 100 + 500 = 1000$ watts. The sum of the connected loads of all the consumers is the connected load to the power station.

(ii) Maximum demand : *It is the greatest demand of load on the power station during a given period.*

The load on the power station varies from time to time. The maximum of all the demands that have occurred during a given period (say a day) is the maximum demand. Thus referring back to the load curve of Fig. 3.2, the maximum demand on the power station during the day is 6 MW and it occurs at 6 P.M. Maximum demand is generally less than the connected load because all the consumers do not switch on their connected load to the system at a time. The knowledge of maximum demand is very important as it helps in determining the installed capacity of the station. The station must be capable of meeting the maximum demand.



Maximum demand meter

(iii) Demand factor. *It is the ratio of maximum demand on the power station to its connected load i.e.,*

$$\text{Demand factor} = \frac{\text{Maximum demand}}{\text{Connected load}}$$

The value of demand factor is usually less than 1. It is expected because maximum demand on the power station is generally less than the connected load. If the maximum demand on the power station is 80 MW and the connected load is 100 MW, then demand factor = $80/100 = 0.8$. The knowledge of demand factor is vital in determining the capacity of the plant equipment.

(iv) Average load. *The average of loads occurring on the power station in a given period (day or month or year) is known as **average load** or **average demand**.*



Energy meter

$$\text{Daily average load} = \frac{\text{No. of units (kWh) generated in a day}}{24 \text{ hours}}$$

$$\text{Monthly average load} = \frac{\text{No. of units (kWh) generated in a month}}{\text{Number of hours in a month}}$$

$$\text{Yearly average load} = \frac{\text{No. of units (kWh) generated in a year}}{8760 \text{ hours}}$$

(v) Load factor. *The ratio of average load to the maximum demand during a given period is known as **load factor** i.e.,*

$$\text{Load factor} = \frac{\text{Average load}}{\text{Max. demand}}$$

If the plant is in operation for T hours,

$$\begin{aligned}\text{Load factor} &= \frac{\text{Average load} \times T}{\text{Max. demand} \times T} \\ &= \frac{\text{Units generated in T hours}}{\text{Max. demand} \times T \text{ hours}}\end{aligned}$$

The load factor may be daily load factor, monthly load factor or annual load factor if the time period considered is a day or month or year. Load factor is always less than 1 because average load is smaller than the maximum demand. The load factor plays key role in determining the overall cost per unit generated. Higher the load factor of the power station, lesser* will be the cost per unit generated.

(vi) Diversity factor. *The ratio of the sum of individual maximum demands to the maximum demand on power station is known as **diversity factor** i.e.,*

$$\text{Diversity factor} = \frac{\text{Sum of individual max. demands}}{\text{Max. demand on power station}}$$

A power station supplies load to various types of consumers whose maximum demands generally do not occur at the same time. Therefore, the maximum demand on the power station is always less than the sum of individual maximum demands of the consumers. Obviously, diversity† factor will always be greater than 1. The greater the diversity factor, the lesser‡ is the cost of generation of power.

(vii) Plant capacity factor. *It is the ratio of actual energy produced to the maximum possible energy that could have been produced during a given period i.e.,*

$$\begin{aligned}\text{Plant capacity factor} &= \frac{\text{Actual energy produced}}{\text{Max. energy that could have been produced}} \\ &= \frac{\text{Average demand} \times T^{**}}{\text{Plant capacity} \times T} \\ &= \frac{\text{Average demand}}{\text{Plant capacity}}\end{aligned}$$

$$\text{Annual plant capacity factor} = \frac{\text{Annual kWh output}}{\text{Plant capacity} \times 8760}$$

The plant capacity factor is an indication of the reserve capacity of the plant. A power station is so designed that it has some reserve capacity for meeting the increased load demand in future. Therefore, the installed capacity of the plant is always somewhat greater than the maximum demand on the plant.

$$\text{Reserve capacity} = \text{Plant capacity} - \text{Max. demand}$$

It is interesting to note that difference between load factor and plant capacity factor is an indication of reserve capacity. If the maximum demand on the plant is equal to the plant capacity, then load factor and plant capacity factor will have the same value. In such a case, the plant will have no reserve capacity.

(viii) Plant use factor. *It is ratio of kWh generated to the product of plant capacity and the number of hours for which the plant was in operation i.e.*

$$\text{Plant use factor} = \frac{\text{Station output in kWh}}{\text{Plant capacity} \times \text{Hours of use}}$$

Suppose a plant having installed capacity of 20 MW produces annual output of 7.35×10^6 kWh and remains in operation for 2190 hours in a year. Then,

$$\text{Plant use factor} = \frac{7.35 \times 10^6}{(20 \times 10^3) \times 2190} = 0.167 = 16.7\%$$

Units Generated per Annum

It is often required to find the kWh generated per annum from maximum demand and load factor.
The procedure is as follows :

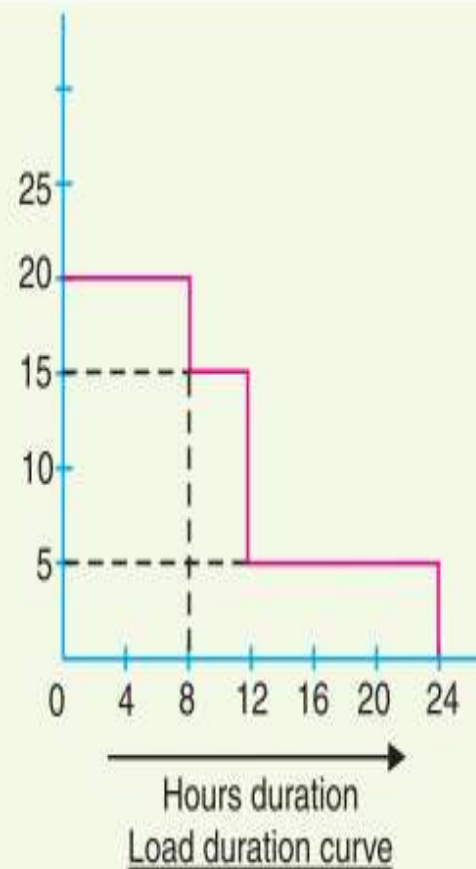
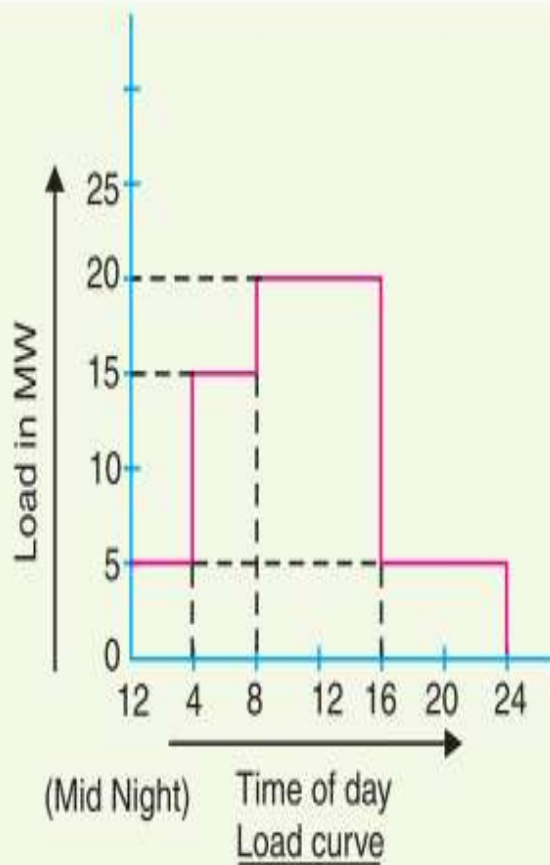
$$\text{Load factor} = \frac{\text{Average load}}{\text{Max. demand}}$$

$$\therefore \text{Average load} = \text{Max. demand} \times \text{L.F.}$$

$$\begin{aligned} \text{Units generated/annum} &= \text{Average load (in kW)} \times \text{Hours in a year} \\ &= \text{Max. demand (in kW)} \times \text{L.F.} \times 8760 \end{aligned}$$

Load Duration Curve

When the load elements of a load curve are arranged in the order of descending magnitudes, the curve thus obtained is called a **load duration curve**.



(i) Domestic load. Domestic load consists of lights, fans, refrigerators, heaters, television, small motors for pumping water etc. Most of the residential load occurs only for some hours during the day (*i.e.*, 24 hours) *e.g.*, lighting load occurs during night time and domestic appliance load occurs for only a few hours. For this reason, the load factor is low (10% to 12%).

(ii) Commercial load. Commercial load consists of lighting for shops, fans and electric appliances used in restaurants etc. This class of load occurs for more hours during the day as compared to the domestic load. The commercial load has seasonal variations due to the extensive use of air-conditioners and space heaters.

(iii) Industrial load. Industrial load consists of load demand by industries. The magnitude of industrial load depends upon the type of industry. Thus small scale industry requires load upto 25 kW, medium scale industry between 25kW and 100 kW and large-scale industry requires load above 500 kW. Industrial loads are generally not weather dependent.

(iv) Municipal load. Municipal load consists of street lighting, power required for water supply and drainage purposes. Street lighting load is practically constant throughout the hours of the night. For water supply, water is pumped to overhead tanks by pumps driven by electric motors. Pumping is carried out during the off-peak period, usually occurring during the night. This helps to improve the load factor of the power system.

(v) Irrigation load. This type of load is the electric power needed for pumps driven by motors to supply water to fields. Generally this type of load is supplied for 12 hours during night.

(vi) Traction load. This type of load includes tram cars, trolley buses, railways etc. This class of load has wide variation. During the morning hour, it reaches peak value because people have to go to their work place. After morning hours, the load starts decreasing and again rises during evening since the people start coming to their homes.

Typical Demand Factor

Type of consumer		Demand factor
<i>Residence lighting</i>	$\frac{1}{4}$ kW	1·00
	$\frac{1}{2}$ kW	0·60
	Over 1 kW	0·50
<i>Commercial lighting</i>	Restaurants	0·70
	Theatres	0·60
	Hotels	0·50
	Schools	0·55
	Small industry	0·60
	Store	0·70
<i>General power service</i>	0–10 H.P.	0·75
	10–20 H.P.	0·65
	20–100 H.P.	0·55
	Over 100 H.P.	0·50

Typical Diversity Factors

	<i>Residential lighting</i>	<i>Commercial lighting</i>	<i>General power supply</i>
Between consumers	3–4	1·5	1·5
Between transformers	1·3	1·3	1·3
Between feeders	1·2	1·2	1·2
Between substations	1·1	1·1	1·1

Load Curves & Selection of Generating Units

The load on a power station is seldom constant; it varies from time to time. Obviously, a single generating unit (*i.e.*, alternator) will not be an economical proposition to meet this varying load. It is because a single unit will have very poor* efficiency during the periods of light loads on the power station. Therefore, in actual practice, a number of generating units of different sizes are installed in a power station. The selection of the number and sizes of the units is decided from the annual load curve of the station. *The number and size of the units are selected in such a way that they correctly fit the station load curve.* Once this underlying principle is adhered to, it becomes possible to operate the generating units at or near the point of maximum efficiency.

Selection of Units

The principle of selection of number and sizes of generating units with the help of load curve is illustrated in Fig. 3.11. In Fig. 3.11 (i), the annual load curve of the station is shown. It is clear from the curve that load on the station has wide variations ; the minimum load being somewhat near 50 kW and maximum load reaching the value of 500 kW. It hardly needs any mention that use of a single unit to meet this varying load will be highly uneconomical.

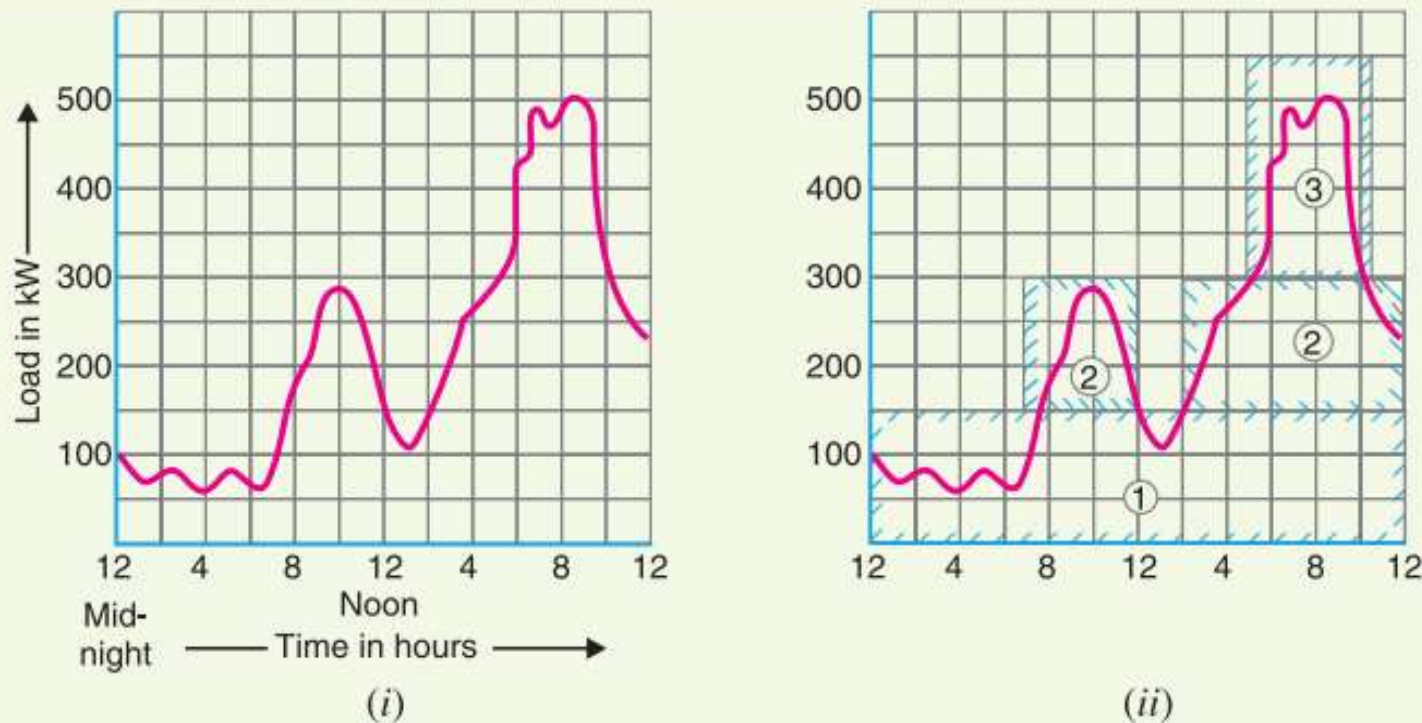


Fig. 3.11

Time

From 12 midnight to 7 A.M.

From 7 A.M. to 12.00 noon

From 12.00 noon to 2 P.M.

From 2 P.M. to 5 P.M.

From 5 P.M. to 10.30 P.M.

From 10.30 P.M. to 12.00 midnight

Units in operation

Only unit no.1 is put in operation.

Unit no. 2 is also started so that both units 1 and 2 are in operation.

Unit no. 2 is stopped and only unit 1 operates.

Unit no. 2 is again started. Now units 1 and 2 are in operation.

Units 1, 2 and 3 are put in operation.

Units 1 and 2 are put in operation.

Selection of Units-Important Points

- (i) The number and sizes of the units should be so selected that they approximately fit the annual load curve of the station.

- (ii) The units should be *preferably* of different capacities to meet the load requirements. Although use of identical units (*i.e.*, having same capacity) ensures saving* in cost, they often do not meet the load requirement.
- (iii) The capacity of the plant should be made 15% to 20% more than the maximum demand to meet the future load requirements.
- (iv) There should be a spare generating unit so that repairs and overhauling of the working units can be carried out.
- (v) The tendency to select a large number of units of smaller capacity in order to fit the load curve very accurately should be avoided. It is because the investment cost per kW of capacity increases as the size of the units decreases.

Base Load & Peak Load Power Stations

The changing load on the power station makes its load curve of variable nature. Fig. 3.13. shows the typical load curve of a power station. It is clear that load on the power station varies from time to time. However, a close look at the load curve reveals that load on the power station can be considered in two parts, namely;

(i) Base load

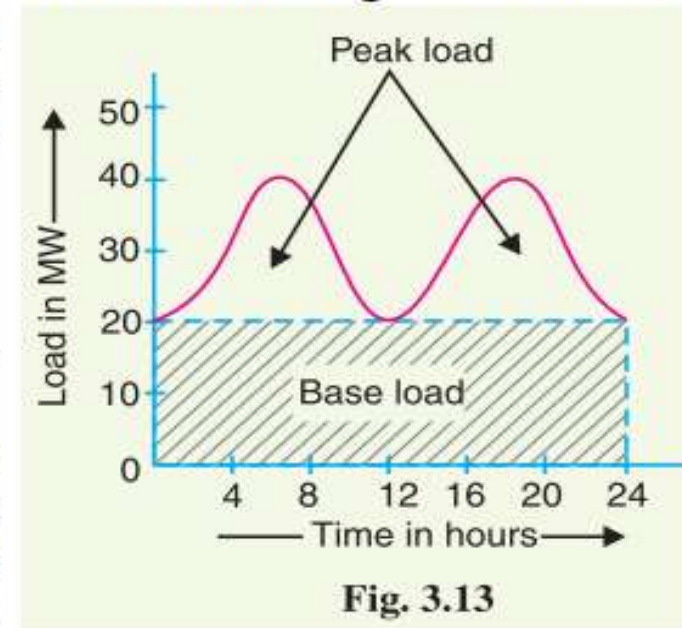
(ii) Peak load

(i) **Base load.** *The unvarying load which occurs almost the whole day on the station is known as **base load**.*

Referring to the load curve of Fig. 3.13, it is clear that 20 MW of load has to be supplied by the station at all times of day and night *i.e.* throughout 24 hours. Therefore, 20 MW is the base load of the station. As base load on the station is almost of constant nature, therefore, it can be suitably supplied (as discussed in the next Article) without facing the problems of variable load.

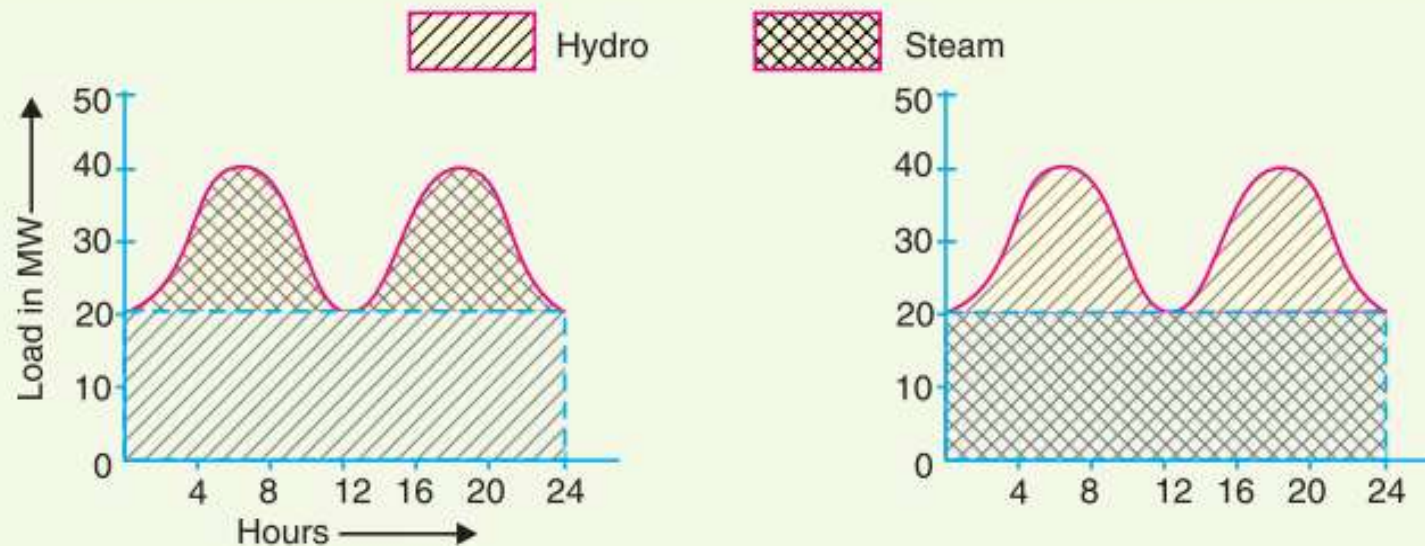
(ii) **Peak load.** *The various peak demands of load over and above the base load of the station is known as **peak load**.*

Referring to the load curve of Fig. 3.13, it is clear that there are peak demands of load excluding base load. These peak demands of the station generally form a small part of the total load and may occur throughout the day.



Illustration

The interconnection of steam and hydro plants is a beautiful illustration to meet the load. When water is available in sufficient quantity as in summer and rainy season, the hydro-electric plant is used to carry the base load and the steam plant supplies the peak load as shown in Fig 3.14 (i).



Load division on Hydro-steam system.

(i) High Flow day

(ii) Low Flow day

Fig. 3.14

However, when the water is not available in sufficient quantity as in winter, the steam plant carries the base load, whereas the hydro-electric plant carries the peak load as shown in Fig. 3.14 (ii).